DATA STRUCTURES

LECTURE 01 (INRODUCTION TO CLASSES)

COURSE OBJECTIVE

- To learn efficient storage mechanisms of data for an efficient access
- To Design and Implement various basic and advanced Data Structures.
- To learn various techniques for representation of the data in the memory.
- To develop applications using appropriate data structures
- To improve the logical ability

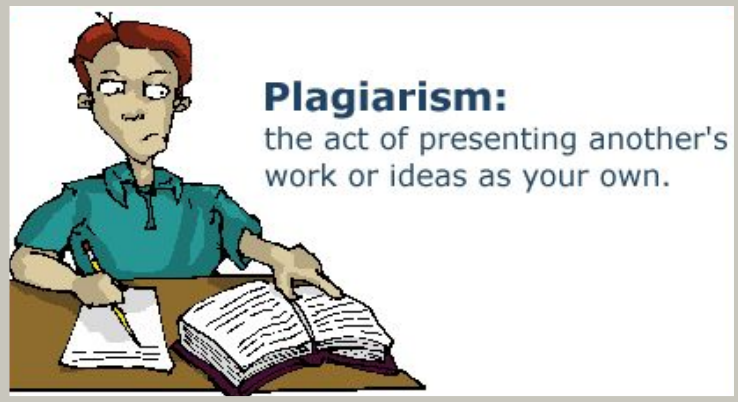
COURSE/ LAB MARKING

- Pre-Lab Tasks (Assignments)
- Lab Tasks (Lab Quiz)
- Post Lab Tasks
- Class Quiz
- Open Lab
- Project
- Sessional I & II
- Final Exam

*Plagiarized stuff will not be appreciated and results in deduction of marks

POLICIES

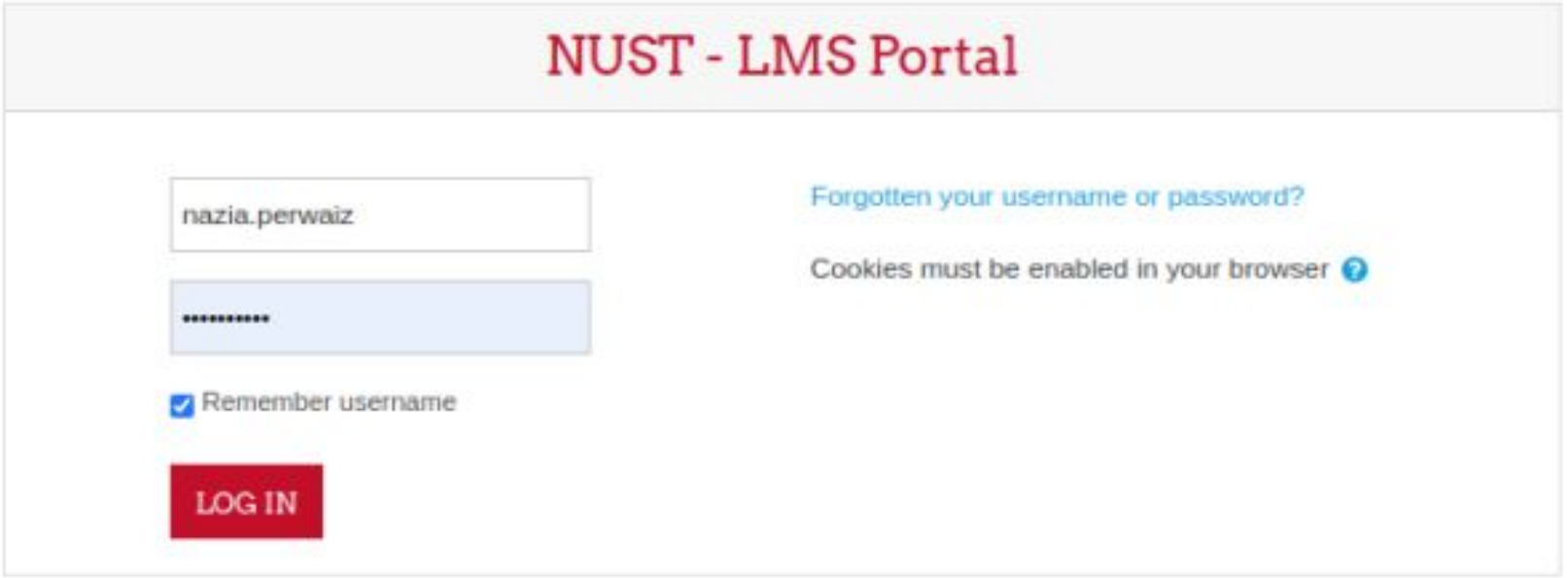
- No extensions will be granted in deadlines.
- Quizzes can be announced/unannounced.
- Exams will be closed book.
- Never cheat.
  - "Better fail NOW or else will fail somewhere LATER in life"
- Plagiarism will also have strict penalties. i.e. will be marked 0
- Reach in time for Class / Lab.



Adapted from: What is Plagiarism? PowerPoint
http://mciu.org/~spjvweb/plagiarism.ppt

ENROLL YOURSELF

1. Login to LMS portal (https://lms.nust.edu.pk)



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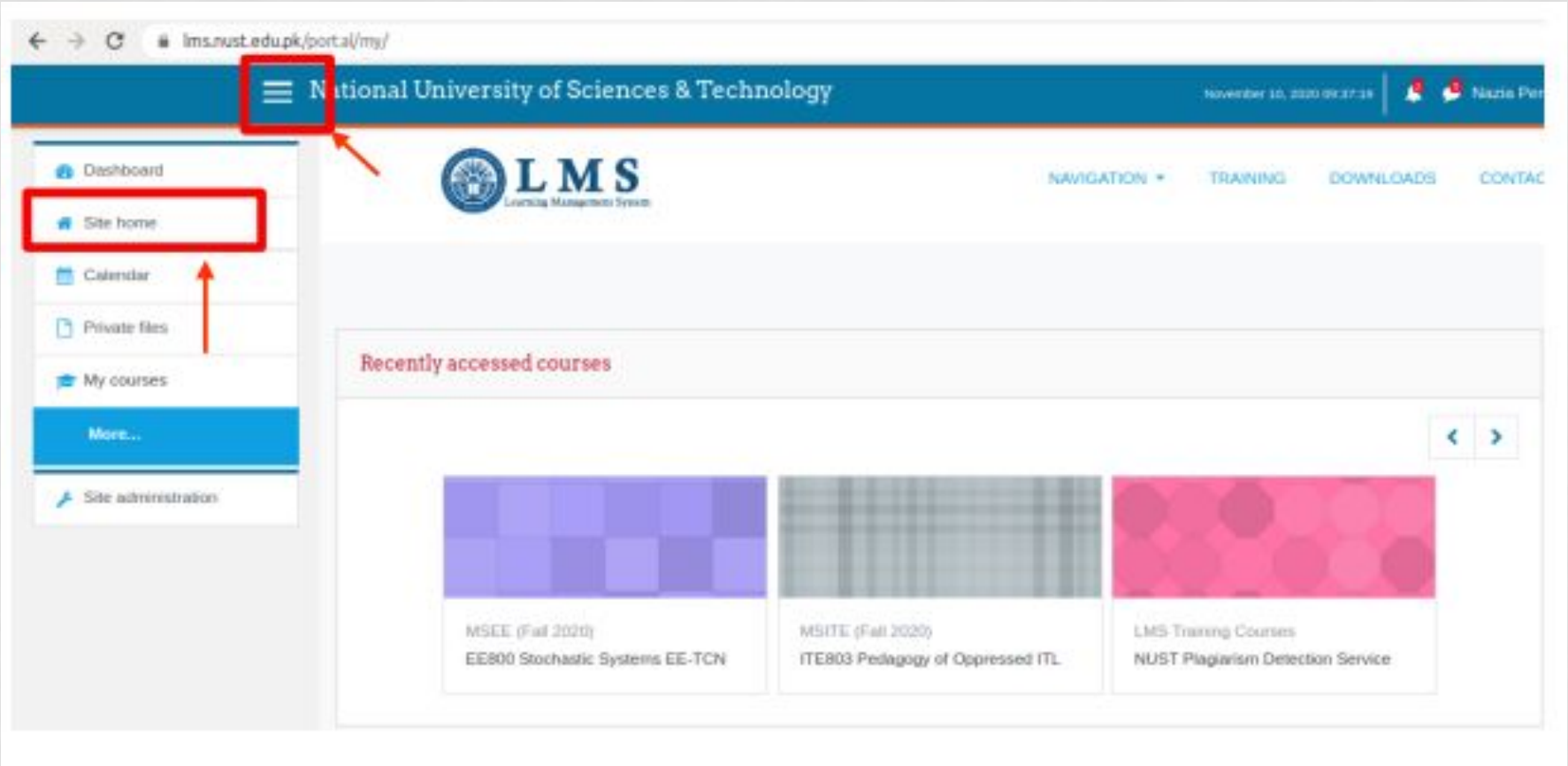

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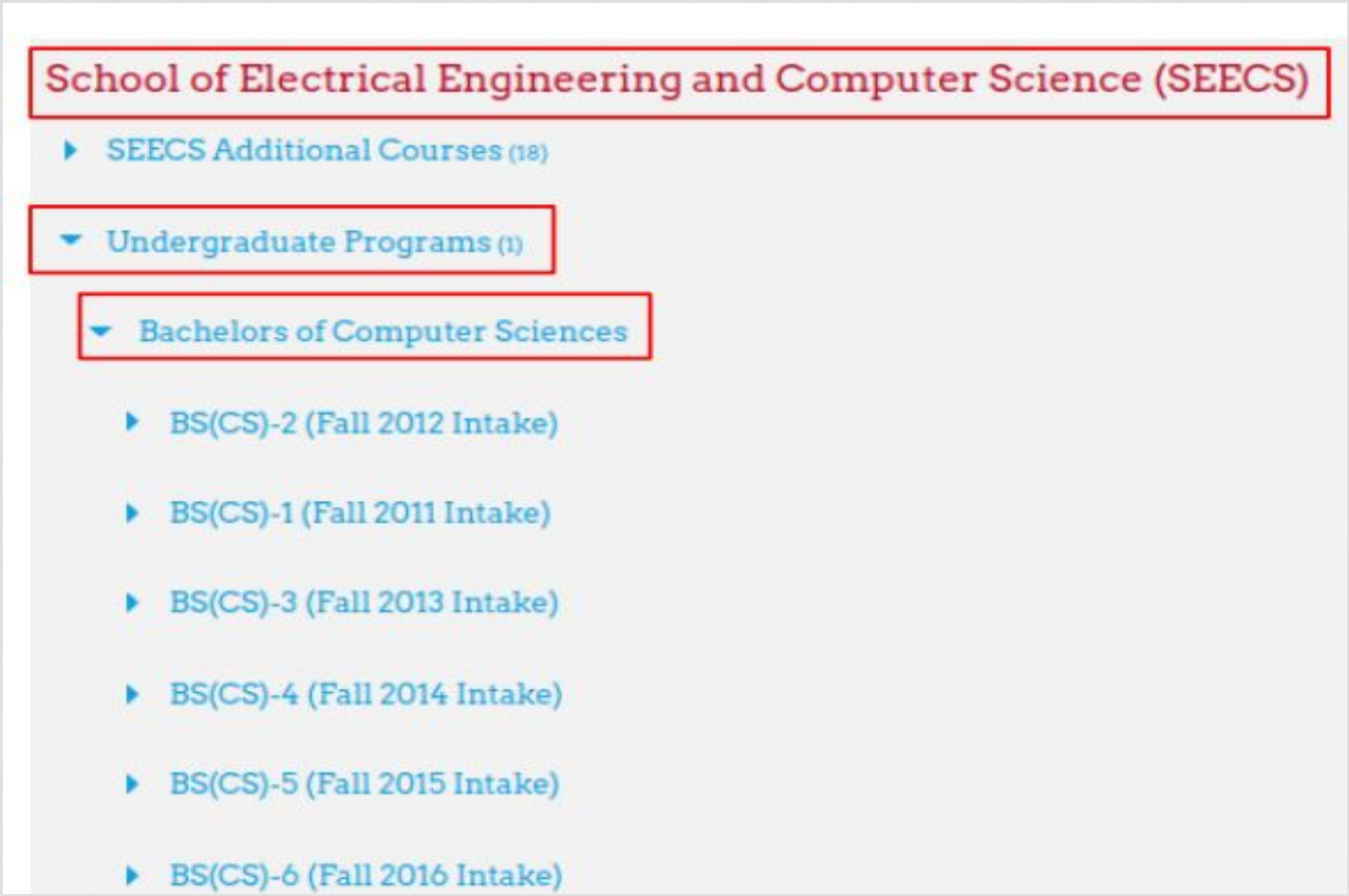

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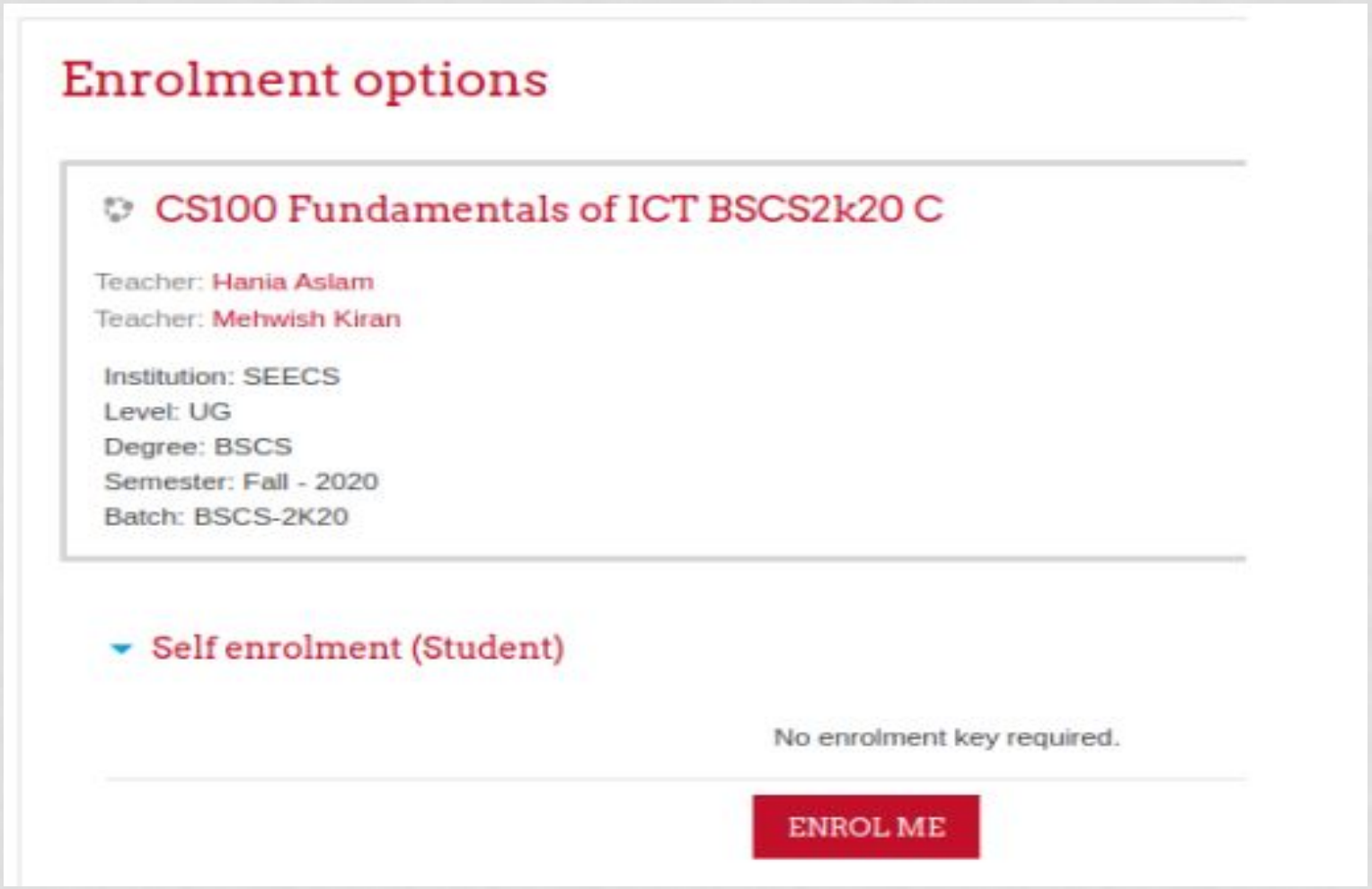

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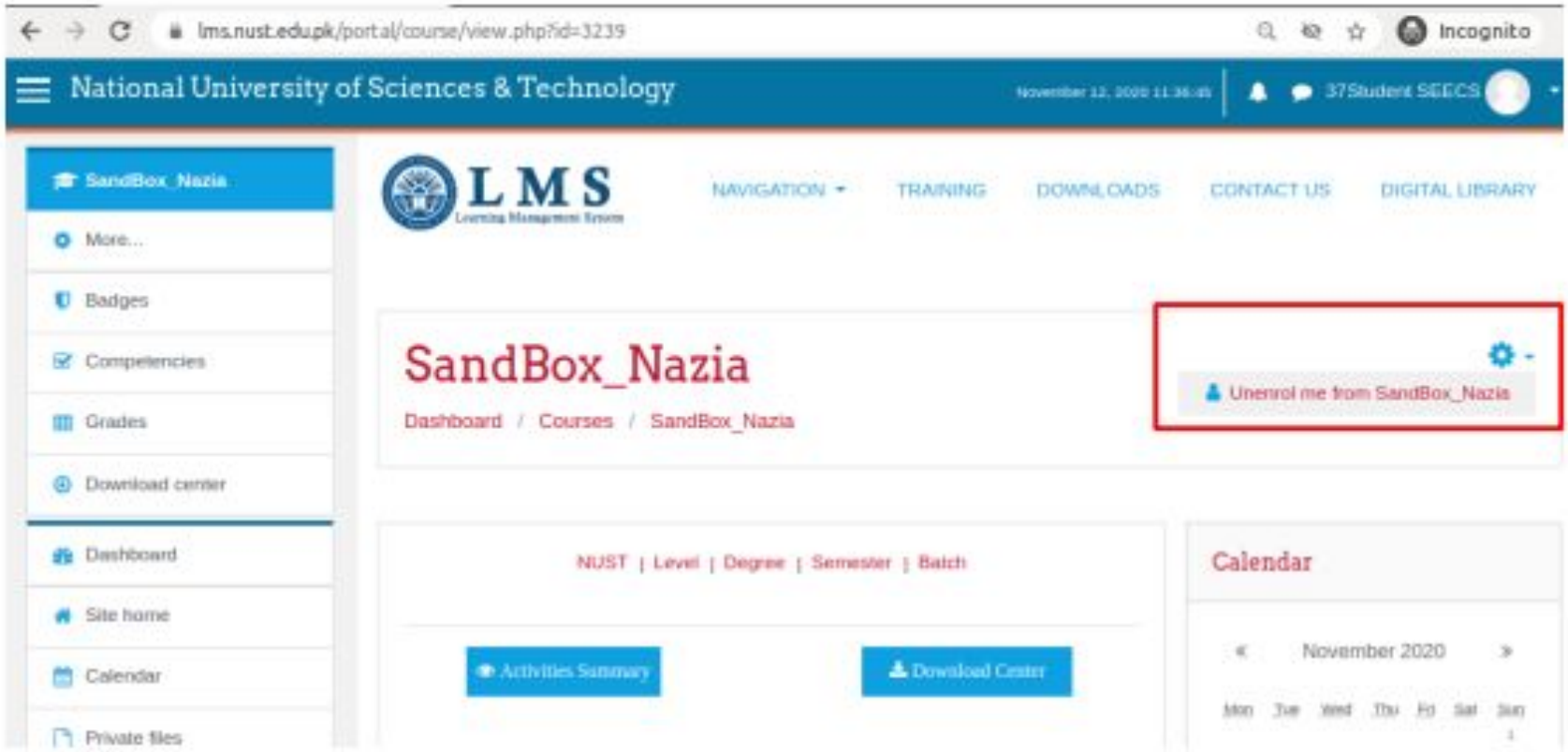

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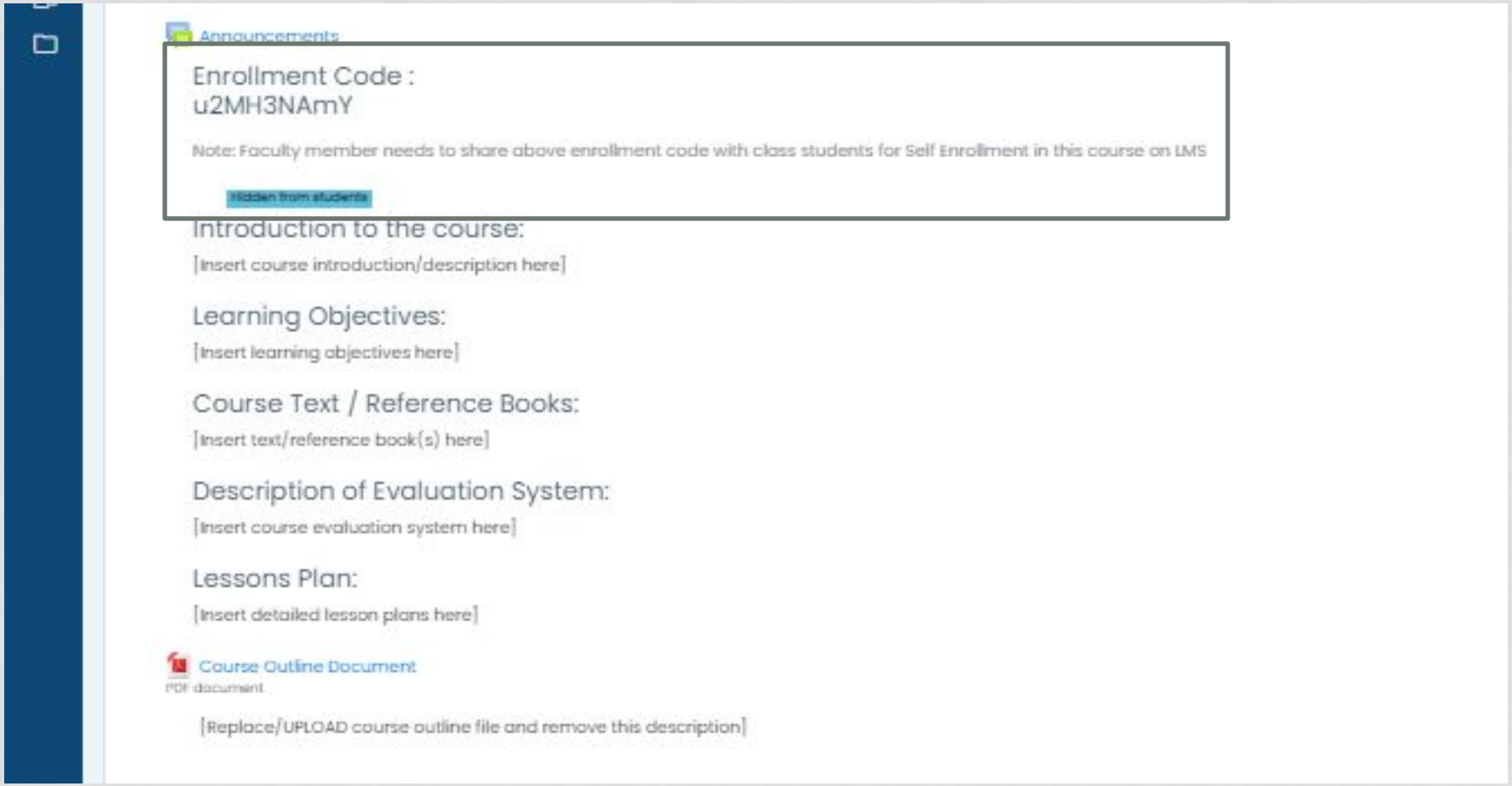


BOOKS

- **Text book**
  - Data Abstraction & Problem solving using C++ by Frank M. Carrano, Timothy Henry, 6th Ed.
  - Data Structures and Algorithm Analysis in C++ by Mark Allen Weiss, 4th Ed.
- **Reference Books**
  - Data Structures & Algorithms in C++ by Michael T. Goodrich, Roberto Tamassia & David Mount, 2nd Ed.

OFFICE HOURS

Office # 01, Faculty Cabin, First Floor, Department of Computer & Software Engineering,

Day	Timings
Monday	1130 -1300 hrs
Tuesday	1130 – 1300 hrs
Friday	1100 – 1230 hrs

COURSE LEARNING OUTCOMES

Course Learning Outcome (CLOs)		PLOs
CLO1	Analyse C++ programs to determine bugs like dangling pointer, bad pointer, memory leakage and shallow copy etc. without using Compiler.	PLO2
CLO2	Develop C++ programs to learn the use of linear and non-linear data structures and design relevant algorithms for them: (stack, queue, dynamically linked lists, trees, graphs, heap, priority queue, hash tables, sorting algorithms, min-max algorithm).	PLO1
CLO3	Analyse algorithms and data structures for performance comparison in terms of time and space complexity using Asymptotic Analysis.	PLO2
CLO4	Solve open ended complex engineering problems by applying appropriate data structures .	PLO3
CLO5	Develop a project that solves a real problem in an application domain (Networks, virtual hard disk, Data mining, machine learning etc.) by using data structures learnt in this course.	PLO3

“HARD WORK BEATS TALENT WHEN TALENT DOESN'T WORK HARD”

-TIM NOTKE

```
while(noSuccess)
{
        tryAgain();

        if(Dead)
                break;
}
```

INTRODUCTION TO DATA STRUCTURES

Digital Data

Movies

Music

Photos

Protein Shapes

DNA

```
gatcttttta tttaaacgat ctctttatta gatctcttat taggatcatg atcctctgtg
gataagtgat tattcacatg gcagatcata taattaagga ggatcgtttg ttgtgagtga
ccggtgatcg tattcgtgat aagctgggat ctaaatggca tgttatgcac agtcactagg
cagaatcaag gttgttatgt ggatatctac tggttttacc ctgcttttaa gcatagttat
acacattcgt tcgcgcgatc tttgagctaa ttagagtaaa ttaatccaat ctttgaccca
```

Maps

INTRODUCTION TO DATA STRUCTURES

Data Structures -> Data StructurING

How do we organize information so that we can find, update, add, and delete portions of it efficiently?

DATA STRUCTURE EXAMPLE APPLICATIONS

- How does Google quickly find web pages that contain a search term?
- How does your Operating System track which memory (disk or RAM) is free?
- How does the processor handles multitasking

WHAT IS A DATA STRUCTURE ANYWAY?

- It's an agreement about:
  - how to store a collection of objects in memory,
  - what operations we can perform on that data,
  - the algorithms for those operations, and
  - how time and space efficient those algorithms are.

HAVE YOU EVER USED A DATA STRUCTURE TO STORE YOUR DATA?

- Arrays in C/C++
  - Stores objects sequentially in memory
  - Can access, change, insert or delete objects
  - Algorithms for insert & delete will shift items as needed

C++ VS. C#

C++	C#
Partially object oriented	Completely Object oriented
Main is a global function	Main is a member function of class
Class members having same Access specifiers requires one time declaration	Access specifier must be mentioned before every class member
Destructors are required to deallocate memory	There is less need for destructors, since dynamically allocated memory will get removed automatically
Platform independent	Targeted towards windows OS
Requires a semicolon after the closing brace at the end of a class definition	Do not requires a semicolon after the closing brace at the end of a class definition

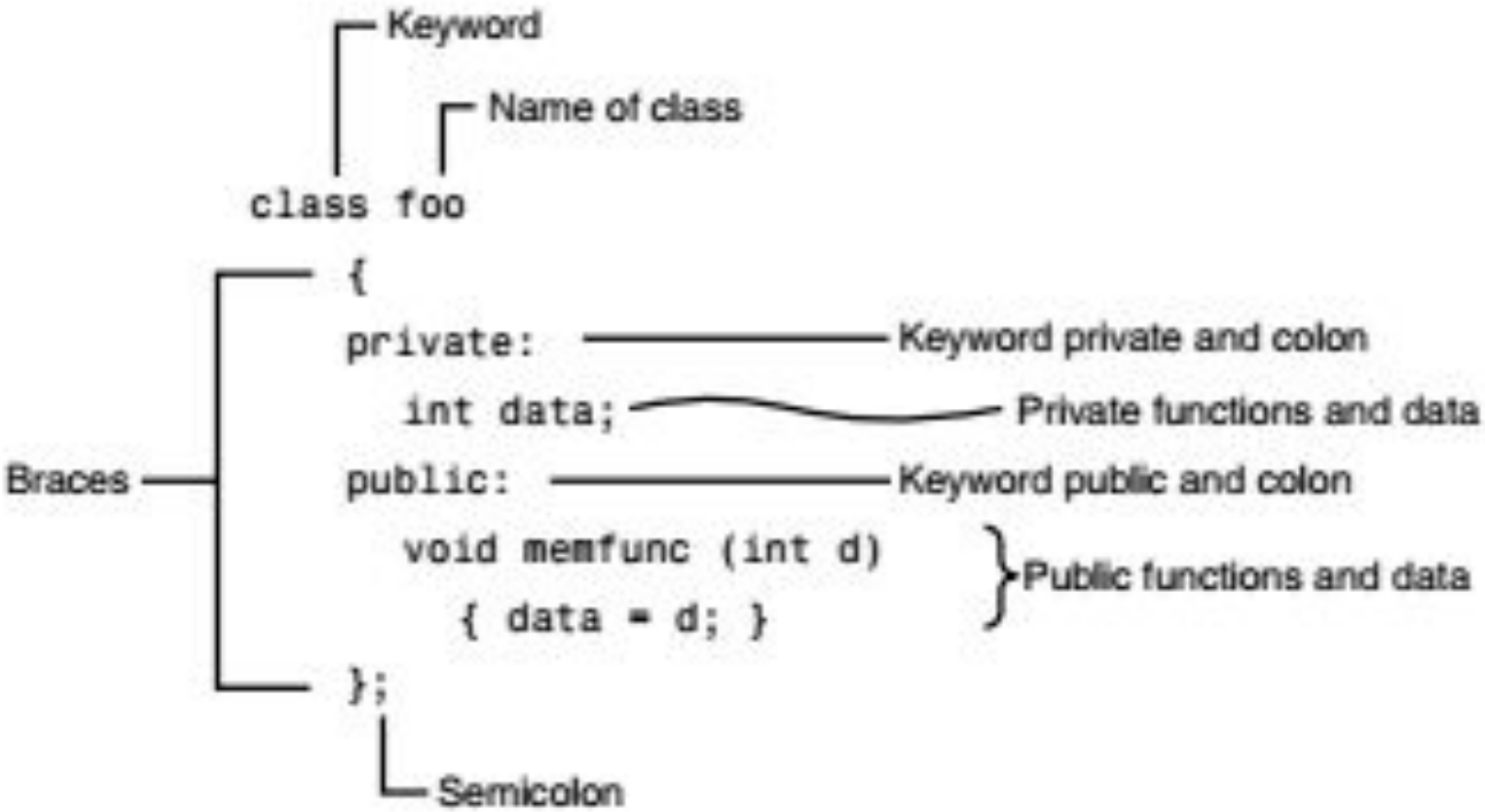
CLASS (ABSTRACT DATA TYPE)

- To define a data type which is a composite of more than one compiler defined data types
- A class serves only as a plan, or a template, or sketch- of a number of similar things (i.e. objects)
- A class is a mechanism for creating user-defined data types.

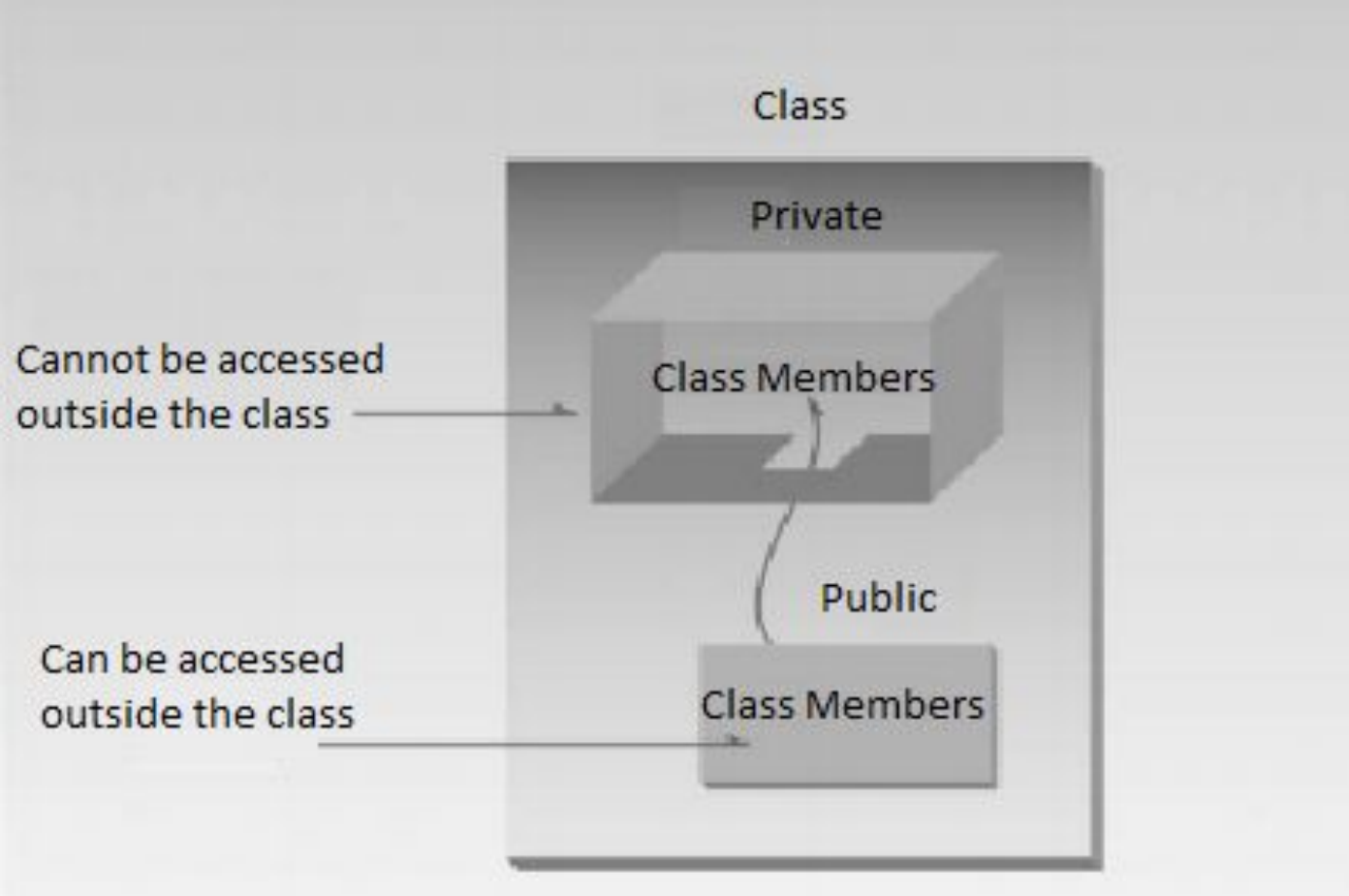
DEFINING A CLASS IN C++

```
class class_name
{
private:
// private class members

public:
// public class members

};
```

STRUCTURE OF A CLASS



WHY WE NEED ACCESS SPECIFIERS



OBJECTS

- A class and an object of that class has the same relationship as a data type and a variable
- All objects with the same characteristics constitute one class.

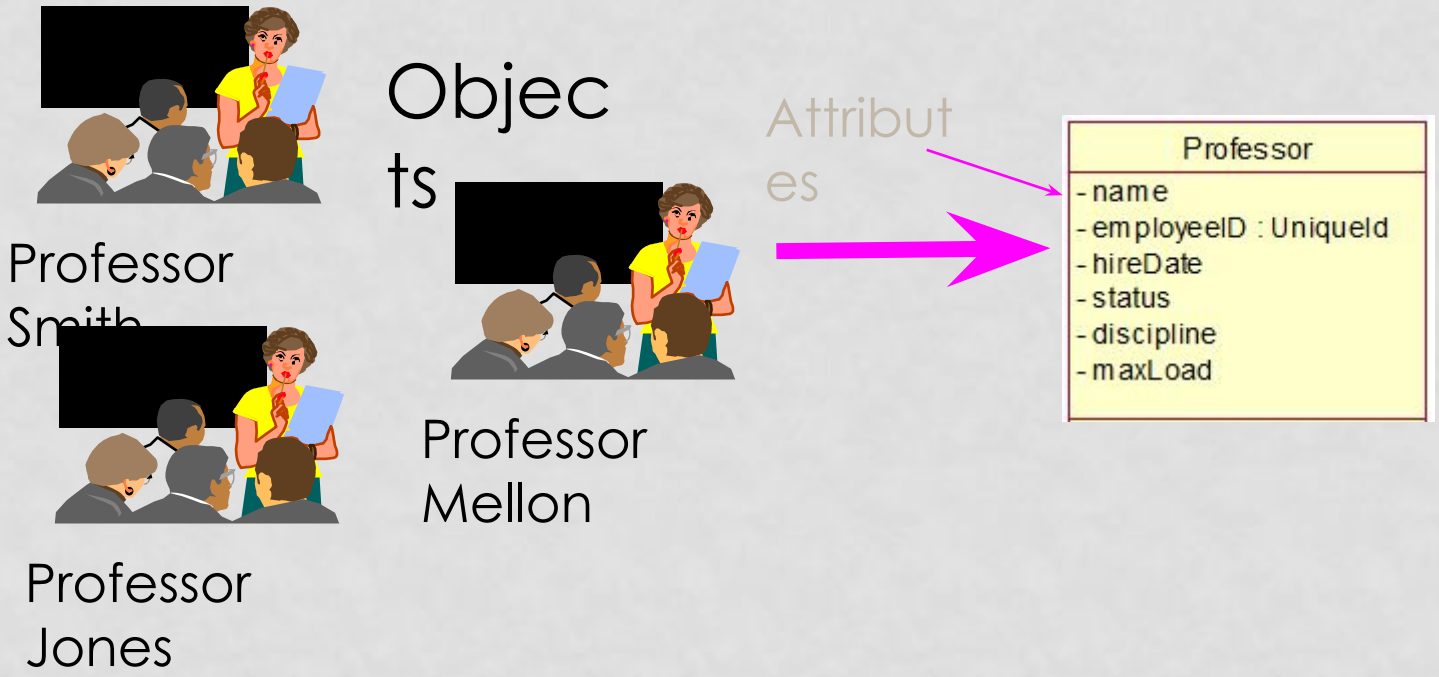
OBJECT INSTANTIATION

Once you create a class type, you can declare one or more objects of that class type. For example:

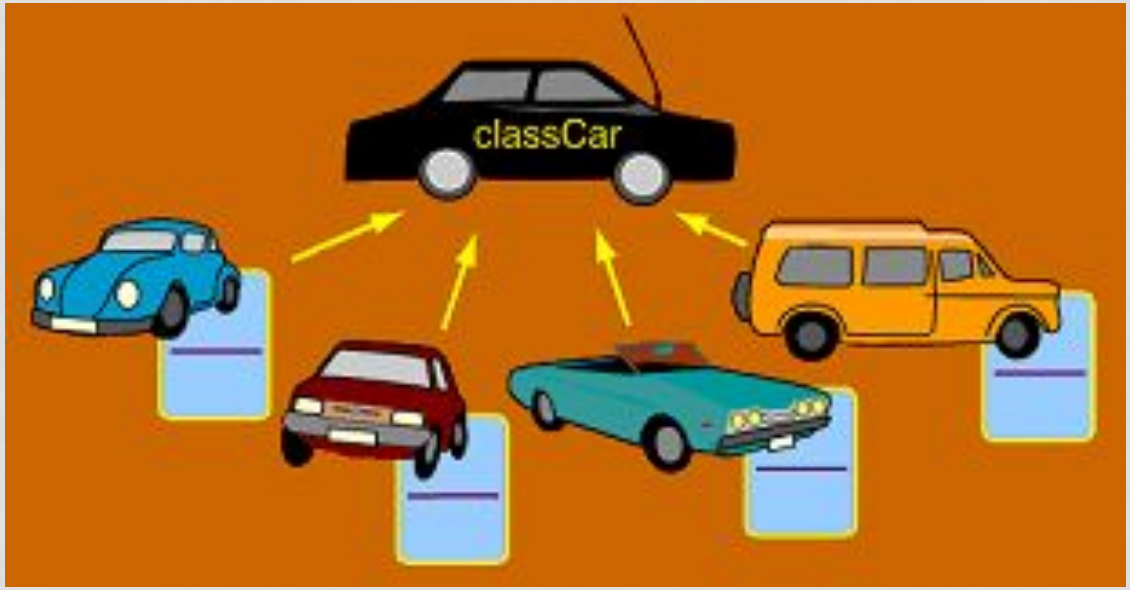
```
class circle{
// class body
}
Void main()
{
circle c;
}
```

OBJECTS ?

- An object is an instance of a class.

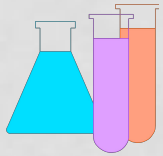


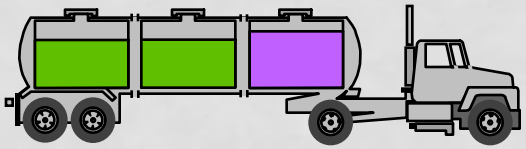
CONTD.



CONTD.

- Informally, an object represents an entity, either physical, conceptual, or software.
  - Physical entity





Truck

  - Conceptual entity

Chemical Process

  - Software entity

Linked List

class circle

```
{
private:
float radius;
public:
circle(){radius=5.5;}
circle(float r) {set(r);}
void set(float r)
{
If(r>0)
{radius = r;}
else
{radius=5.5;}
}
```

```
float calcArea()
{
return 3.14*power(radius,2);
}
float getRadius()
{
Return radius;
} };
 void main()
 {
 circle c1; //create circles
 circle c2(6);circle c3(-3);
 c1.calcArea();
 cout<<c3.calcArea()<<endl;
  cout<<c3.getRadius();
}
```

DEFAULT PARAMETERIZED CONSTRUCTOR

```
Class Circle
{
Float rad;
Circle(float r=5.5)
{
rad=r;
}
};
```

class circle

```
{
private:
float radius;
public:
circle(float r=5.5)
{
Set(r);
}
void set(float r)
{
If(r>0)
{radius = r;}
else
{radius=5.5;}
}
float calcArea()
{
return 3.14*power(radius,2);
}
float getRadius()
{
Return radius;
}};
 void main()
 {
 circle c1; //create circles
 circle c2(6); circle c3(-3);
 c1.calcArea();
 cout<<c3.calcArea()<<endl;
  cout<<c3.getRadius();
}
```

CALL FOR CONSTRUCTOR AND DESTRUCTOR

- When a variable of programmer defined data type is created and constructor is called it will result in an automatic call to constructor of all the variables of programmer defined data type present inside i.e. when a circle type variable is created constructor is called for class circle.

- Similarly whenever scope of variable ends, destructor of the class is called.

CLASS TASK

- Make a class **"Cube"**. Choose appropriate **data members.** Your program should be able to calculate:

  - **Surface Area of cube (formula area= $6a^2$)**
  - **Volume of cube (volume = a^3)**

FAQS

- MS Teams
  Queries

- LMS
  Lab Task submission
  Assignment submission
  Project submission

REFERNCE

- Chapter 16 , C How to Program by Deitel & Deitel, 9th Ed.